

## SLUBICE, Poland

WMO: 123100

Lat: 52.348N

Lon: 14.593E

Elev: 79

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 9.3        | 15.6 | 4.4                           | 6.9 | 12.0 | 10.0 | 9.2 | 19.9 | 20.8                  | 43.4 | 18.8 | 41.6 | 3.9                   | 110  | 0.513 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.1                   | 87.6            | 67.2 | 84.2 | 66.3 | 80.9 | 65.2 | 70.1                | 81.4 | 68.4 | 79.4 | 66.9 | 77.2 | 6.9                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.6                            | 98.1 | 73.7 | 64.8 | 92.2 | 71.6 | 63.3 | 87.3 | 70.5 | 34.1          | 81.3 | 32.7 | 79.1 | 31.5 | 77.2 | 75.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 17.0          | 14.9 | 13.4 | 3.4                        | 93.8 | 8.5                | 3.7 | -2.7                                               | 96.5 | -7.7       | 98.6 | -12.4      | 100.7 | -18.6      | 103.4 |    |    |
| (5)               |      |      | 3.2                        | 73.5 | 8.2                | 1.6 | -2.7                                               | 74.6 | -7.4       | 75.5 | -12.0      | 76.4  | -18.0      | 77.6  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 49.7   | 31.9  | 33.5  | 40.1  | 49.7  | 57.1  | 63.6  | 67.1  | 66.1  | 58.4  | 49.6  | 41.6  | 36.3  |  |  |
|                                                                        | DBStd       | 14.20  | 9.62  | 8.34  | 7.92  | 7.36  | 7.04  | 6.22  | 5.52  | 5.39  | 5.67  | 7.02  | 6.61  | 7.70  |  |  |
|                                                                        | HDD50       | 2232   | 560   | 464   | 316   | 93    | 16    | 0     | 0     | 0     | 3     | 95    | 261   | 424   |  |  |
|                                                                        | HDD65       | 5872   | 1025  | 883   | 773   | 459   | 261   | 99    | 42    | 50    | 212   | 477   | 703   | 888   |  |  |
|                                                                        | CDD50       | 2110   | 0     | 1     | 7     | 85    | 236   | 407   | 529   | 500   | 254   | 83    | 7     | 1     |  |  |
|                                                                        | CDD65       | 278    | 0     | 0     | 0     | 1     | 17    | 56    | 106   | 85    | 13    | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 2976   | 0     | 0     | 0     | 26    | 245   | 660   | 1095  | 784   | 163   | 3     | 0     | 0     |  |  |
|                                                                        | CDH80       | 894    | 0     | 0     | 0     | 4     | 57    | 193   | 360   | 246   | 34    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 6.5    | 7.4   | 6.9   | 7.3   | 6.8   | 6.3   | 6.2   | 5.6   | 5.6   | 5.6   | 6.0   | 6.5   | 7.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 20.1   | 1.3   | 1.0   | 1.0   | 1.3   | 2.2   | 2.2   | 2.2   | 2.1   | 1.6   | 1.7   | 1.8   | 1.6   |  |  |
|                                                                        | PrecMax     | 26.6   | 3.7   | 2.2   | 1.9   | 2.5   | 5.5   | 4.4   | 5.2   | 4.0   | 3.5   | 4.2   | 2.9   | 4.1   |  |  |
|                                                                        | PrecMin     | 14.6   | 0.4   | 0.1   | 0.3   | 0.2   | 0.7   | 0.3   | 0.2   | 0.2   | 0.2   | 0.1   | 0.6   | 0.1   |  |  |
|                                                                        | PrecStd     | 3.7    | 0.8   | 0.6   | 0.5   | 0.6   | 1.1   | 1.3   | 1.5   | 1.3   | 0.8   | 1.2   | 0.7   | 1.1   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 54.0  | 56.4  | 66.2  | 77.3  | 85.9  | 90.5  | 92.3  | 92.4  | 84.3  | 72.8  | 59.0  | 54.8  |  |  |
|                                                                        |             | MCWB   | 48.6  | 48.3  | 52.6  | 58.5  | 64.4  | 68.7  | 69.0  | 68.1  | 65.7  | 60.2  | 53.9  | 49.8  |  |  |
|                                                                        | 2%          | DB     | 49.9  | 51.0  | 60.3  | 72.0  | 80.9  | 85.0  | 87.5  | 86.8  | 78.9  | 67.7  | 55.5  | 51.1  |  |  |
|                                                                        |             | MCWB   | 47.1  | 45.6  | 49.9  | 56.3  | 63.1  | 66.4  | 67.4  | 67.6  | 63.2  | 58.3  | 51.4  | 47.8  |  |  |
|                                                                        | 5%          | DB     | 46.9  | 47.9  | 56.1  | 68.4  | 75.7  | 81.3  | 84.2  | 81.7  | 74.2  | 63.9  | 53.1  | 48.8  |  |  |
|                                                                        |             | MCWB   | 43.9  | 43.2  | 47.8  | 53.9  | 60.4  | 65.0  | 66.7  | 66.2  | 61.9  | 57.1  | 49.9  | 45.7  |  |  |
|                                                                        | 10%         | DB     | 44.3  | 45.1  | 52.3  | 64.5  | 71.5  | 77.3  | 80.4  | 78.0  | 69.9  | 60.4  | 51.1  | 46.4  |  |  |
|                                                                        |             | MCWB   | 41.7  | 41.4  | 45.8  | 52.7  | 58.2  | 63.4  | 65.6  | 64.9  | 59.9  | 55.4  | 48.3  | 43.8  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 50.2  | 50.4  | 54.4  | 60.7  | 67.3  | 71.6  | 72.3  | 72.2  | 68.0  | 62.4  | 55.0  | 51.1  |  |  |
|                                                                        |             | MCDB   | 52.4  | 53.7  | 63.0  | 74.5  | 79.8  | 84.2  | 84.0  | 84.0  | 80.0  | 69.8  | 58.2  | 53.4  |  |  |
|                                                                        | 2%          | WB     | 47.2  | 46.7  | 51.3  | 57.8  | 64.7  | 68.7  | 70.1  | 69.4  | 64.9  | 59.7  | 52.3  | 48.1  |  |  |
|                                                                        |             | MCDB   | 49.7  | 49.7  | 58.6  | 68.8  | 76.5  | 80.1  | 81.7  | 80.4  | 75.6  | 65.6  | 54.7  | 50.4  |  |  |
|                                                                        | 5%          | WB     | 44.2  | 43.7  | 48.7  | 55.7  | 62.5  | 66.4  | 68.6  | 67.8  | 62.9  | 57.7  | 50.3  | 46.0  |  |  |
|                                                                        |             | MCDB   | 46.6  | 47.4  | 54.4  | 66.2  | 72.6  | 77.6  | 79.5  | 78.3  | 71.2  | 62.9  | 52.8  | 48.6  |  |  |
|                                                                        | 10%         | WB     | 42.0  | 41.3  | 46.5  | 53.4  | 60.3  | 64.6  | 67.1  | 66.2  | 61.1  | 55.7  | 48.5  | 43.8  |  |  |
|                                                                        |             | MCDB   | 44.1  | 45.0  | 52.0  | 62.3  | 69.4  | 74.3  | 77.2  | 75.2  | 68.8  | 60.0  | 50.8  | 46.1  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.8   | 11.1  | 15.1  | 19.8  | 19.8  | 19.9  | 19.1  | 19.3  | 18.2  | 14.5  | 9.6   | 8.0   |  |  |
|                                                                        |             | MCDBR  | 9.7   | 15.7  | 23.8  | 27.1  | 27.3  | 27.3  | 27.1  | 26.6  | 26.3  | 19.8  | 11.8  | 9.8   |  |  |
|                                                                        |             | MCWBR  | 7.8   | 10.6  | 14.5  | 13.7  | 12.0  | 11.7  | 10.3  | 11.2  | 12.6  | 11.9  | 8.6   | 7.7   |  |  |
|                                                                        | 5% WB       | MCDBR  | 9.5   | 13.5  | 20.3  | 23.7  | 23.5  | 24.0  | 22.5  | 22.3  | 22.9  | 17.7  | 11.4  | 9.4   |  |  |
|                                                                        |             | MCWBR  | 8.0   | 9.9   | 13.1  | 12.6  | 11.7  | 11.4  | 9.8   | 10.5  | 12.3  | 11.3  | 8.6   | 8.0   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.324 | 0.351 | 0.390 | 0.420 | 0.417 | 0.414 | 0.435 | 0.429 | 0.389 | 0.375 | 0.351 | 0.318 |  |  |
|                                                                        | taud        |        | 2.313 | 2.290 | 2.205 | 2.194 | 2.250 | 2.286 | 2.238 | 2.268 | 2.353 | 2.350 | 2.353 | 2.331 |  |  |
|                                                                        | Ebn at Noon |        | 209   | 238   | 250   | 257   | 265   | 267   | 258   | 252   | 249   | 226   | 196   | 189   |  |  |
|                                                                        | Edh at Noon |        | 20    | 27    | 36    | 41    | 41    | 40    | 41    | 38    | 31    | 26    | 20    | 17    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 211   | 430   | 771   | 1285  | 1569  | 1665  | 1565  | 1361  | 987   | 546   | 244   | 152   |  |  |
|                                                                        | RadStd      |        | 29    | 76    | 102   | 176   | 183   | 134   | 194   | 127   | 126   | 74    | 37    | 16    |  |  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (38 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page